Reinforcement Learning – COMS4061A/7071A

Introduction to Reinforcement Learning

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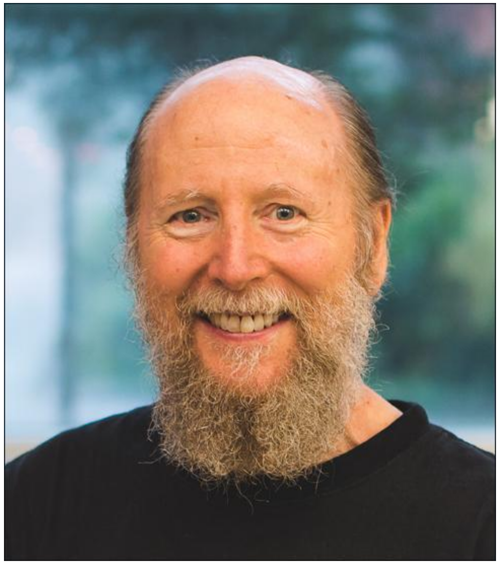
Based on slides by **Benjamin Rosman**, Joelle Pineau, Doina Precup, and **Rich Sutton**

2024 Turing Awards!!!



Richard S. Sutton

Andrew Barto

“For developing the conceptual and algorithmic foundations of Reinforcement Learning”

Course logistics

Course will be run through Moodle page

Lectures:

- Wednesdays **2pm-4pm** in MSL005

- We also have a lab slot: Wednesdays **4pm-5pm** in MSL005
  - Only some weeks

- Many lectures have hand-ins
- I will post slides before the lecture

Assessment

Face-to-face exam at the end of the year

Large group assignment

Four small lab assignments

Weekly hand-ins

Small quizzes

Course outline

1. Introduction to RL
2. Multi-Armed Bandits
3. Markov Decision Processes
4. Dynamic Programming and Value Iteration
5. Temporal Difference Methods
6. Function Approximation
7. Model-Based RL
8. Policy Gradient Methods
9. Hierarchical Reinforcement Learning

Textbook

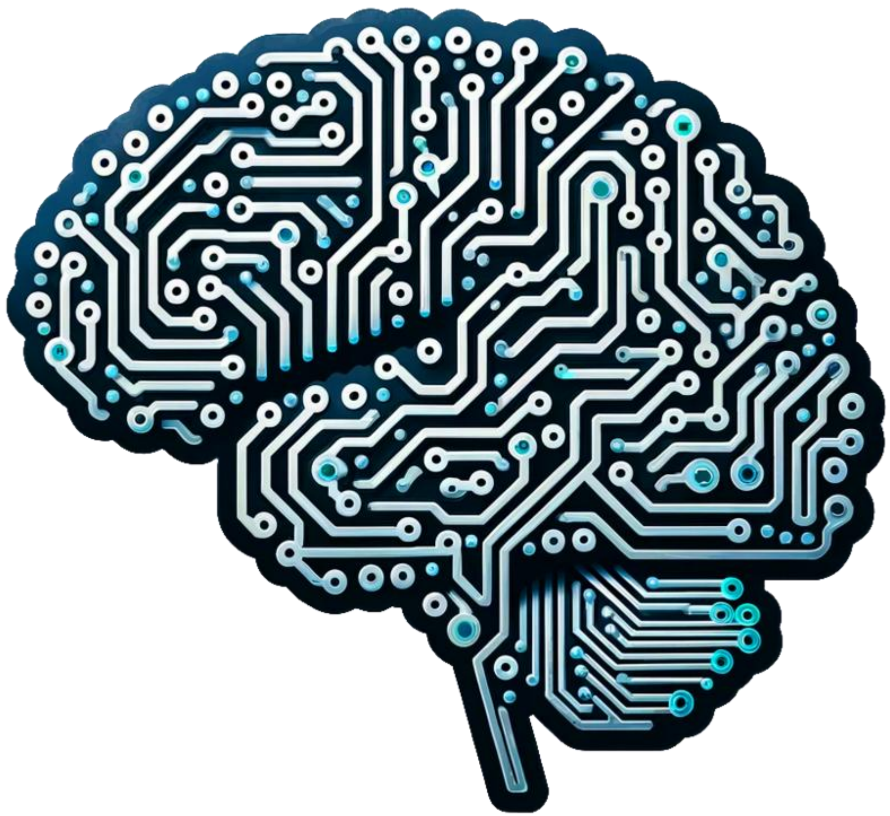
We will be loosely following <u>the</u> RL textbook:

Sutton, Richard S., and Andrew G. Barto. *Reinforcement learning: An introduction*. MIT press, 2018.

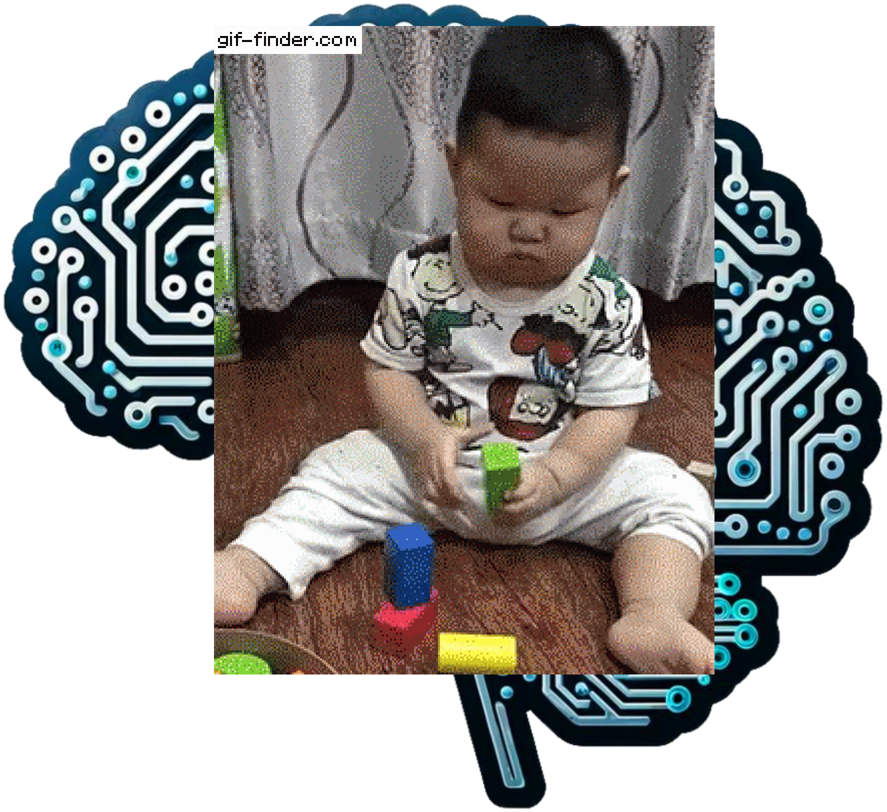
http://www.incompleteideas.net/book/RLbook2020.pdf

Now, on to the fun stuff…

What is intelligence?



What is intelligence?



ML is usually making predictions

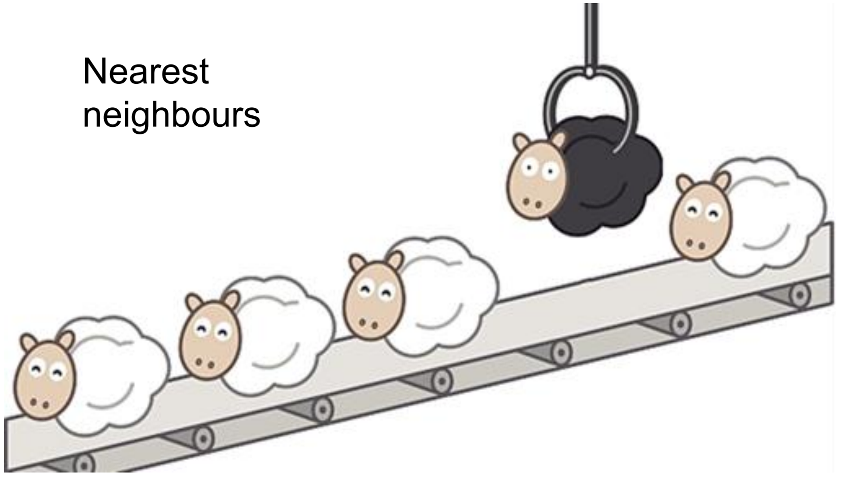
Decision
trees

Logistic
regression

Neural
networks

Nearest
neighbours

Probabilistic graphical
models



Random
forests

Naïve Bayes

Support vector
machines

Why do we make predictions?

Predict the weather?
    Bring an umbrella!

Predict stock prices?
    Buy and sell

Predict disease spread?
    Change public policy

We use predictions to inform our behaviours

Decision making

Intelligence is about acting

How to choose actions?

But:

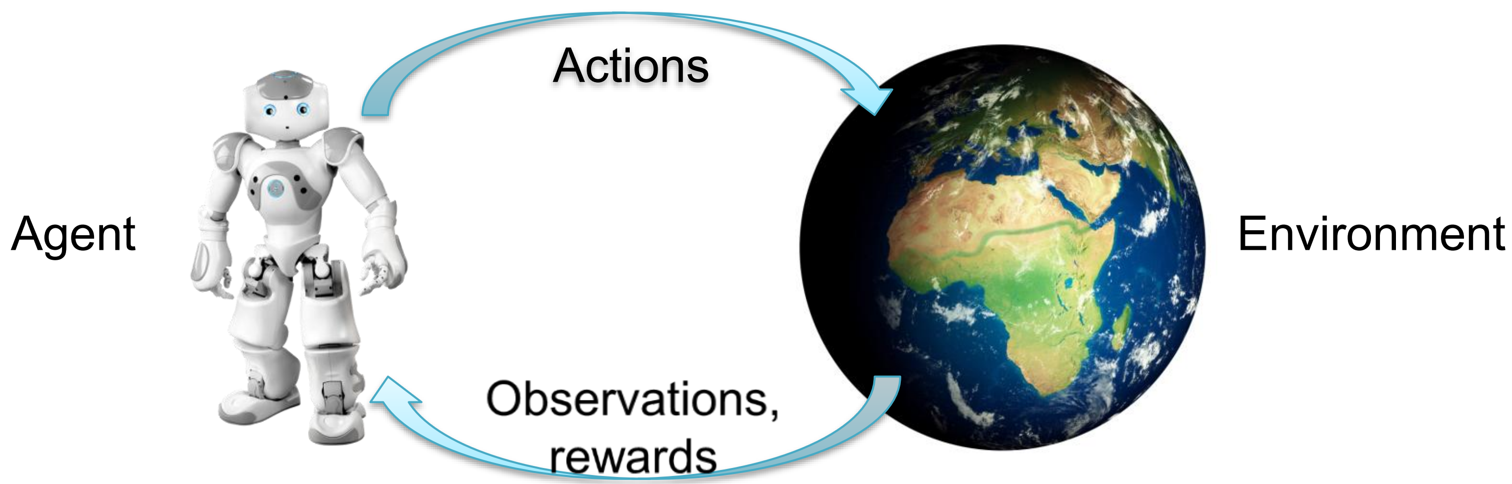
- Actions produce effects
- Different outcomes may be more/less desirable
- Want to think long-term: optimise overall performance

Implications for safety! Why?

Reinforcement learning

Think about the setting of an agent in an environment

- Choose actions to maximise long-term utility/rewards
- Learn by trial-and-error – improve with experience



The RL problem is to find a policy (behaviour): optimal actions for each state of the environment to maximise total rewards

Difference to supervised learning

In SL: an “expert” gives us labels to learn from

- The “right answers”
- Learn input-output function

In RL: some events give us positive/negative “rewards”

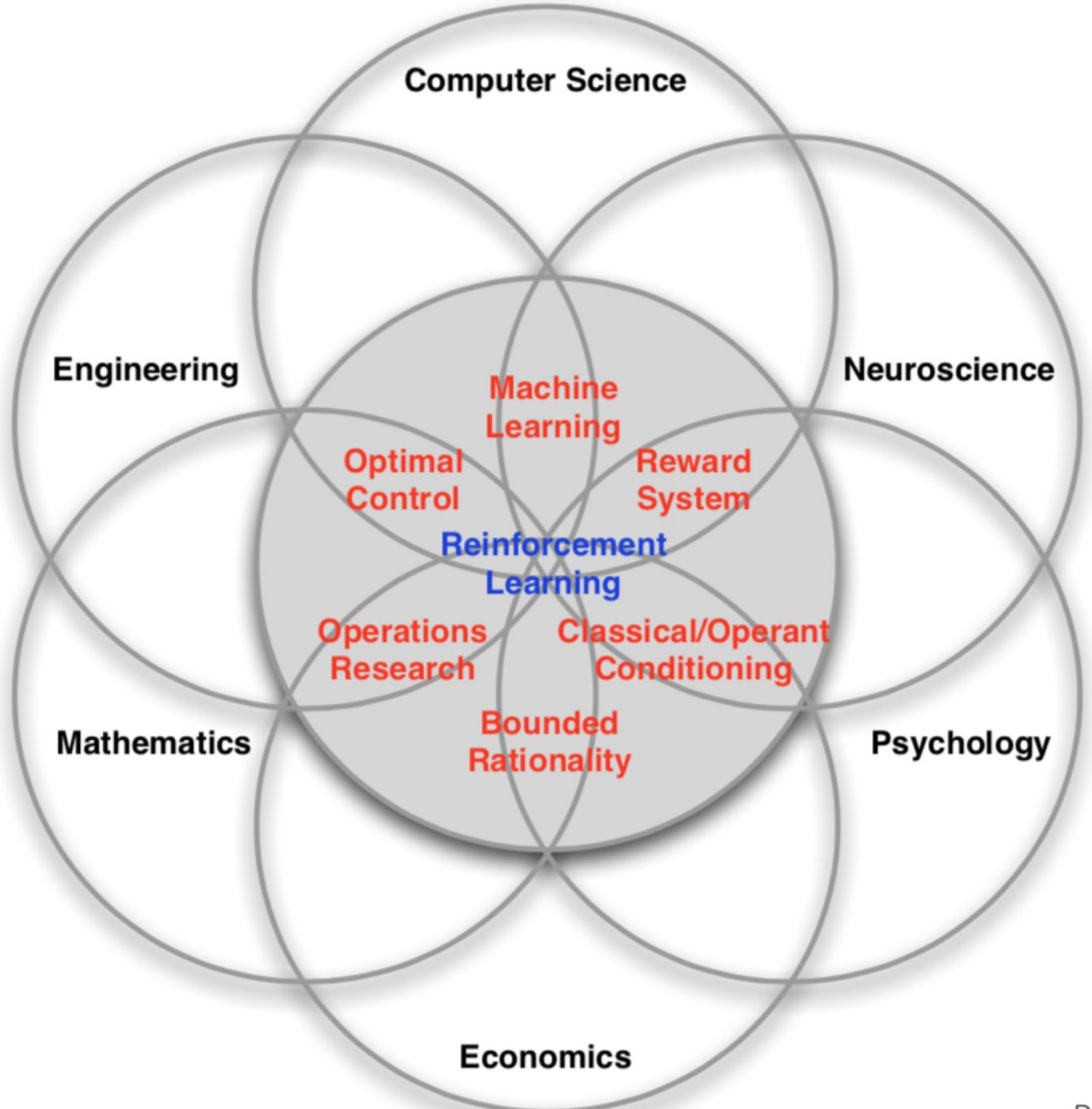
- Numerical feedback
- Often delayed rewards
- Learn behaviours

Encoding what to do, rather than how to do it

Based on ideas from behaviour learning:

- Human/animal psychology
- “Carrots and sticks”
- Dopamine

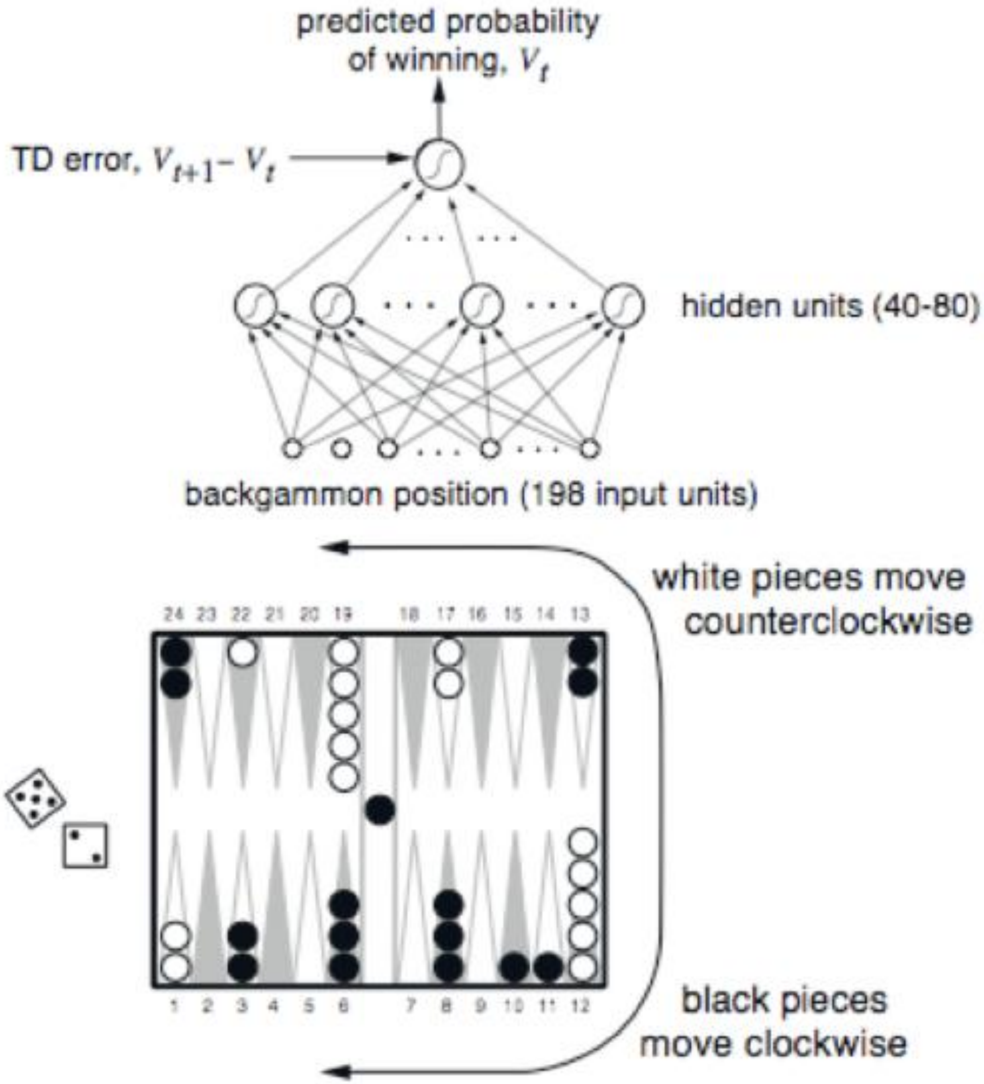
Homework: Watch this talk “**Andy Barto** - In the Beginning ML was RL - RLC 2024”
https://www.youtube.com/watch?v=-gQNM7rAWP0



David Silver 2015

RL successes: TD-Gammon

Tesauro, 1992-95



Reward:
- Win: +100
- Lose: -100
- Else: 0

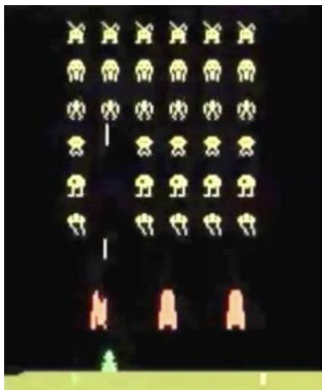
Training: played 1.5×10^6 games against itself

Play at level of best humans

Changed how people play

RL successes: Atari Games

Google Deepmind, 2015

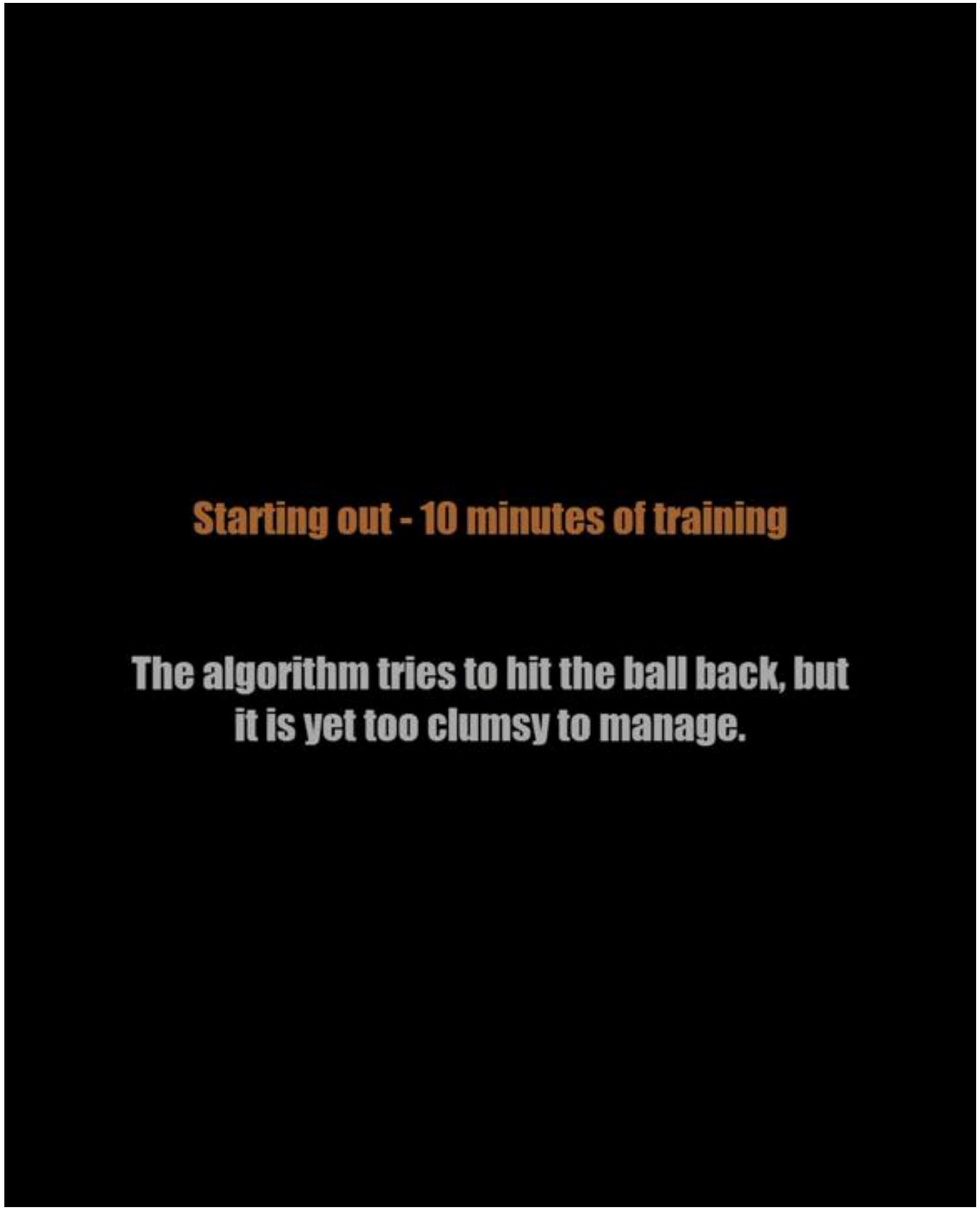


Learned to play 49 Atari games, from self-play

Screen pixels → joystick commands

Same algorithm!

Outperform humans on many of them

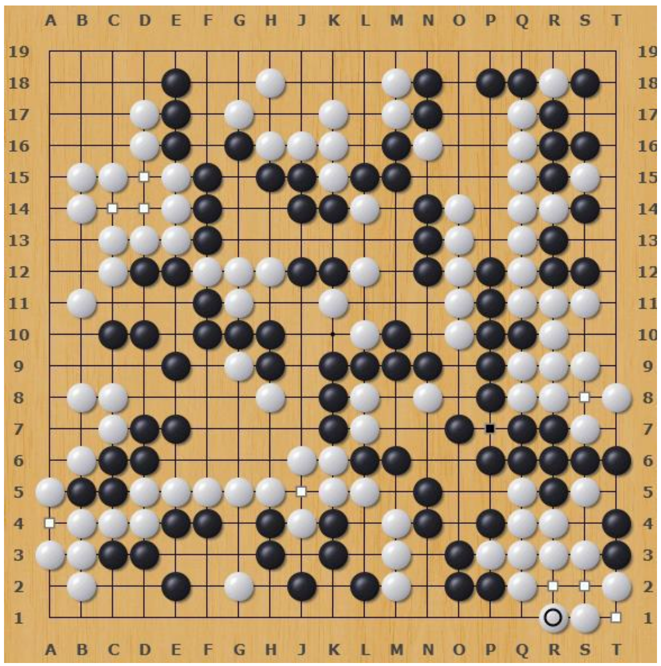


video: Two Minute Papers

Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A.A., Veness, J., Bellemare, M.G., Graves, A., Riedmiller, M., Fidjeland, A.K., Ostrovski, G. and Petersen, S., 2015. Human-level control through deep reinforcement learning. *Nature*, *518*(7540), pp.529-533.

RL successes: AlphaGo

Google Deepmind, 2016



Learned to play Go
- 361 moves
- ~10^{174} board states

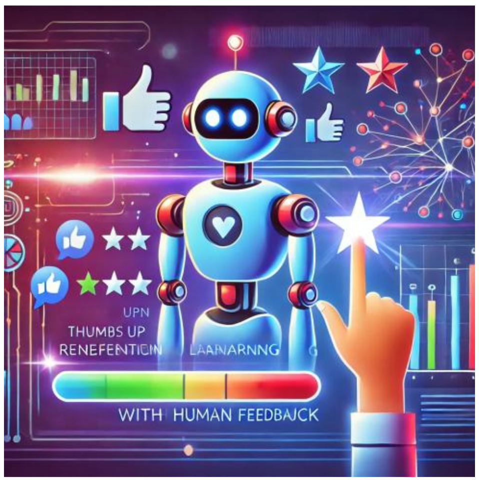
Original version trained on human games
- Later outperformed by self-playing version

Developed new strategies!

Homework: Watch the documentary
https://www.youtube.com/watch?v=WXuK6gekU1Y

RL successes: RLHF

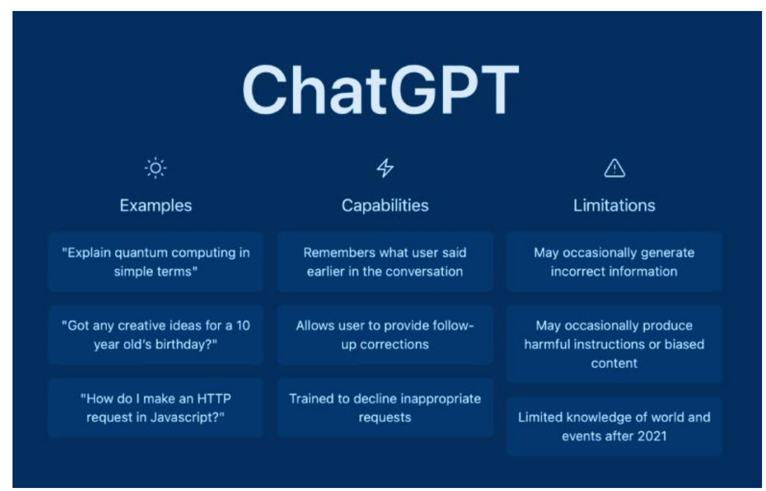
Google & OpenAI, 2017
OpenAI, 2020



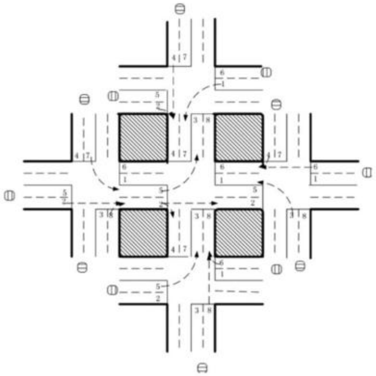
Learn human preferences from human feedback

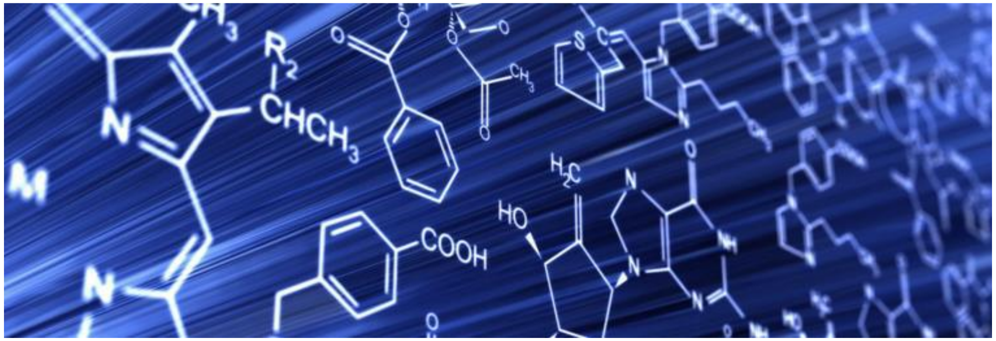
Train a model to predict preferences

Fine-tune LLM to max these preferences



Applications





And more…


PRODUCTION
DeepMind
STARCRAFT
DEMONSTRATION
5:22
REPLAY
Catalyst LE
DeepMind

The plan

In this course, we'll be looking at the kinds of algorithms that go into these problems

Unlike other areas of ML, we usually can't just use pre-existing datasets!
Why?

> Often use simulators instead
> We need to see what happens after we make choices!

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